UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/580,857	01/21/2022	Franco Falconi	2021P00003 US02 (969US2)	2969

114292 7590 03/06/2024

Locke Lord LLP
IP Docket Department
P.O. BOX 55874
Boston, MA 02205

EXAMINER	
EVERETT, CHRISTOPHER E	

ART UNIT	PAPER NUMBER
2116	

NOTIFICATION DATE	DELIVERY MODE
03/06/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

Patent-Docketing@schneider-electric.com
patent@lockelord.com

Office Action Summary	Application No. 17/580,857	Applicant(s) Falconi et al.	
	Examiner CHRISTOPHER E EVERETT	Art Unit 2116	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 1/21/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-18 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-18 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/21/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Claims 1-18 are pending.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

Information Disclosure Statement

The listing of references in the specification is not a proper information disclosure statement. 37 CFR 1.98(b) requires a list of all patents, publications, or other information submitted for consideration by the Office, and MPEP § 609.04(a) states, "the list may not be incorporated into the specification but must be submitted in a separate paper." Therefore, unless the references have been cited by the examiner on form PTO-892, they have not been considered.

Claim Objections

The numbering of claims is not in accordance with 37 CFR 1.75 which requires the claims to be numbered consecutively in Arabic numerals. The claim set filed on 1/21/2022 has two (2) claims labeled as 3.

> 3. The control system of claim 1, wherein the response model includes a linear part and a non-linear part, and the predictive control algorithm employs the response model to remove non-linearity, thereby compensating for time delay and enabling linear control through an RST control strategy.
>
> 3. The control system of claim 1, wherein the closed loop control includes a feedback loop to an output of a command law to make an estimation of an input disturbance.

Misnumbered claims 3-17 been renumbered 4-18. Appropriate correction of the claim numbering and dependencies are required.

Claim 2 is objected to because of the following informalities: The limitation "the coefficients" is recited. There is insufficient antecedent basis for this limitation in the claim. Appropriate correction is required.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Claim 18 (re-numbered) is rejected under 35 U.S.C. 101 because the applicant has provided evidence that the applicant intends the term "tangible computer medium" to include non-statutory matter. The applicant describes a computer-usable media as including open ended language and thus it is reasonable to interpret it to include all possible mediums, including non-statutory mediums (see paragraph 0087). The words "storage" and/or "recording" are insufficient to convey only statutory embodiments to one of ordinary skill in the art absent an explicit and deliberate limiting definition or clear differentiation between storage media and transitory media in the disclosure. As such, the claim is drawn to a form of energy. Energy is not one of the four categories of invention and therefore this claim is not statutory. Energy is not a series of steps or acts and thus is not a process. Energy is not a physical article or object and as such is not a machine or manufacture. Energy is not a combination of substances and therefore not a composition of matter.

The Examiner suggests amending the claim to read as a “non-transitory computer medium”.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claims 1, 4-9, and 12-18 are rejected under 35 U.S.C. 103 as being unpatentable over I. D. Landau, "Robust digital control of systems with time delay (the Smith predictor revisited)," Proceedings of 1994 33rd IEEE Conference on Decision and Control, Lake Buena Vista, FL, USA, 1994, pp. 865-869 vol.1 (Landau) (cited by Applicant) in view of U.S. Patent Application Publication No. 2009/0112335 (Mehta).

Claim 1:

The cited prior art describes a control system for controlling a device or system, comprising: **(Landau: "The paper investigates the design of robust digital controllers for systems with time delay. A particular class of digital controllers is considered. namely those for which the closed loop poles include the poles of the system to he controlled." Abstract; "Digital controllers offers a natural and easy solution for controlling systems with time delay without restrictions upon the complexity of the plant dynamics." Section 1)**

Landau does not explicitly describe memory, processor, or tunable as described below. However, Mehta teaches the memory, processor, and tunable as described below.

- memory; and **(Mehta: "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)**
- at least one processor configured to: **(Mehta: "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)**
    - receive a setpoint signal as input; and **(Landau: see the y*(t+d+1) input as illustrated in figure 2.1b)**
    - perform closed loop control of the device or system by outputting a control signal according to the setpoint signal and a response model of the device or system using a predictive control algorithm, **(Landau: see the y(t) control signal output from the RST controller using an explicit predictor as illustrated in figure 2.1b)**
        - the predictive control algorithm being configured to implement control according to a polynomial representation for regulation, sensitivity and

tracking and **(Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)**

- further implement non-linearity or time delay compensation using the response model, **(Landau: see the predictor as illustrated in figures 2.1a; see also the time delay variations as described in section 6)**
- the closed loop control being tunable using an adjustable single parameter for accelerating or decelerating the closed loop control relative to an open loop control scenario. **(Mehta: "In accordance with some embodiments of the disclosed technique, the function block plug-in 64 performs data collection, as well as model and tuning control to calculate tuning variables for the control function block 58 and process model identification control logic to monitor data and conditions for trigger events indicating a process change sufficient for model identification." Paragraph 0037; "Using the above controller 11 and workstation 13 architecture, the performance of the process may be improved by identifying underperforming loops, tune controllers and/or loops, perform testing, monitor performance. For example, the controllers 11, and specifically the function block plug-in 64 and model identification module 72, utilize learning algorithms to automatically identify new models based on process changes (either automatic, manual or induced) that trigger the controller 11 to collected the control data (e.g., process input data) and operating condition (e.g., process output data) and calculate a new model or model parameters.**

The model or model parameters may be utilized by a user of the workstation 13, and of the application 84 in particular, to evaluate performance, identify actual or potential non-linearities and degradations in the process, identify underperforming processes and abnormal process conditions, and recommend tuning adjustments (e.g., tuning parameters). The virtual controller may be used to test the tuning adjustments to predict process performance and analyze the prediction before implementing the tuning adjustments." Paragraph 0055)

One of ordinary skill in the art would have recognized that applying the known technique of Landau, namely, a digital control system, with the known techniques of Mehta, namely, intelligent control and monitoring in a process control system, would have yielded predictable results and resulted in an improved system. Accordingly, applying the teachings of Landau to use RST with a predictor for a digital control system with the teachings of Mehta to use various optimization techniques for a process control system would have been recognized by those of ordinary skill in the art as resulting in an improved process control system (i.e., using various tuning techniques for a RST with a predictor digital control system of Landau based on the teachings of using tuning techniques for a process control system in Mehta).

Claim 4 (re-numbered):

Landau does not explicitly describe disturbance as described below. However, Mehta teaches the disturbance as described below.

The cited prior art describes the control system of claim 1, wherein the closed loop control includes a feedback loop to an output of a command law to make an estimation of an input disturbance. **(Mehta: "Control routines, and the function blocks that implement such routines, have been configured in accordance with a number of control techniques, including PID control, fuzzy logic control, and model-based techniques such as a Smith Predictor or Model Predictive control (MPC). In model-based control techniques, the parameters used in the routines to determine the closed loop control response are based on the dynamic process response to changes in the manipulated or measured disturbances serving as inputs to the process. A representation of this response of the process to changes in process inputs may be characterized as a process model. For instance, a first-order parameterized process model may specify values for the gain, dead time, and time constant of a self-regulating process or values for integrating gain and dead time of an integrating process." Paragraph 0006; Landau: see the y(t) control signal output from the RST controller using an explicit predictor as illustrated in figure 2.1b)**

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 5 (re-numbered):

Landau does not explicitly describe equipment as described below. However, Mehta teaches the equipment as described below.

The cited prior art describes the control system of claim 1, wherein the device or system comprises equipment to be controlled in an industrial process or system. **(Mehta: "The field devices 15-22 may be any types of devices, such as sensors, valves, transmitters, positioners,**

etc., while the I/O cards 26 and 28 may be any types of I/O devices conforming to any desired communication or controller protocol." Paragraph 0031; "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 6 (re-numbered):

Landau does not explicitly describe dead time as described below. However, Mehta teaches the dead time as described below.

The cited prior art describes the control system of claim 1, wherein the time delay comprises a dead time. **(Mehta: "The process model identification control logic 110 may then invoke the data collection and model identification. Upon receiving the model identification results, the process model identification control logic 110 may analyze the results for allowable ranges for gain, time constant and dead time if the process is a self-regulating process or for integrating gain and dead time if the process is an integrating process, where the allowable ranges may be user configurable. Thereafter, the tuning 114 of the control function block 58 may be set by the model and tuning control logic 106 based on the identified model and selected tuning rules." Paragraph 0065; Landau: see the predictor as illustrated in figures 2.1a; see also the time delay variations as described in section 6)**

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 7 (re-numbered):

Landau does not explicitly describe a sensor as described below. However, Mehta teaches the sensor as described below.

The cited prior art describes the control system of claim 1, further comprising: one or more sensors for sensing operating characteristic(s) of the device or system, which is feedback for closed loop control. **(Mehta: "As illustrated by the exploded block 30 of FIG. 1, the controller 11 may include a number of single-loop control routines, illustrated as routines 32 and 34, and, if desired, may implement one or more advanced control loops, illustrated as control loop 36. Each such loop is typically referred to as a control module. The single-loop control routines 32 and 34 are illustrated as performing single loop control using a single-input/single-output fuzzy logic control block and a single-input/single-output PID control block, respectively, connected to appropriate analog input (AI) and analog output (AO) function blocks, which may be associated with process control devices such as valves, with measurement devices such as temperature and pressure transmitters, or with any other device within the process control system 10. The advanced control loop 36 is illustrated as including an advanced control block 38 having inputs communicatively connected to one or more AI function blocks and outputs communicatively connected to one or more AO function blocks, although the inputs and outputs of the advanced control block 38 may be connected to any other desired function blocks or control elements to receive other types of inputs and to provide other types of control outputs. The advanced**

control block 38 may be any type of model predictive control (MPC) block, neural network modeling or control block, a multi-variable fuzzy logic control block, a real-time-optimizer block, etc. It will be understood that the function blocks illustrated in FIG. 1, including the advanced control block 38, can be executed by the controller 11 or, alternatively, can be located in and executed by any other processing device, such as one of the workstations 13 or even one of the field devices 19-22." Paragraph 0034; "The field devices 15-22 may be any types of devices, such as sensors, valves, transmitters, positioners, etc., while the I/O cards 26 and 28 may be any types of I/O devices conforming to any desired communication or controller protocol." Paragraph 0031; "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 8 (re-numbered):

Landau does not explicitly describe memory or processor as described below. However, Mehta teaches the memory or processor as described below.

The cited prior art describes the control system of claim 1, wherein the memory and at least one processor are part of a controller for an industrial process or system. **(Mehta: "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein**

or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 9 (re-numbered):

Landau does not explicitly describe a computer or tunable as described below. However, Mehta teaches the computer and tunable as described below.

The cited prior art describes a computer-implemented method of controlling a device or system, comprising: **(Landau: "The paper investigates the design of robust digital controllers for systems with time delay. A particular class of digital controllers is considered. namely those for which the closed loop poles include the poles of the system to he controlled." Abstract; "Digital controllers offers a natural and easy solution for controlling systems with time delay without restrictions upon the complexity of the plant dynamics." Section 1; Mehta: "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)**

- receiving a setpoint signal as input; and **(Landau: see the y*(t+d+1) input as illustrated in figure 2.1b)**

- performing closed loop control of the device or system by outputting a control signal according to the setpoint signal and a response model of the device or system using a predictive control algorithm, **(Landau: see the y(t) control signal output from the RST controller using an explicit predictor as illustrated in figure 2.1b)**
    - the predictive control algorithm being configured to implement control according to a polynomial representation for regulation, sensitivity and tracking and **(Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)**
    - further implement non-linearity or time delay compensation using the response model, **(Landau: see the predictor as illustrated in figures 2.1a; see also the time delay variations as described in section 6)**
    - the closed loop control being tunable using an adjustable single parameter for accelerating or decelerating the closed loop control relative to an open loop control scenario. **(Mehta: "In accordance with some embodiments of the disclosed technique, the function block plug-in 64 performs data collection, as well as model and tuning control to calculate tuning variables for the control function block 58 and process model identification control logic to monitor data and conditions for trigger events indicating a process change sufficient for model identification." Paragraph 0037; "Using the above controller 11 and workstation 13 architecture, the performance of the process may be improved by identifying underperforming loops, tune controllers and/or loops, perform testing, monitor performance. For example, the controllers 11, and specifically the function block plug-in 64 and**

model identification module 72, utilize learning algorithms to automatically identify new models based on process changes (either automatic, manual or induced) that trigger the controller 11 to collected the control data (e.g., process input data) and operating condition (e.g., process output data) and calculate a new model or model parameters. The model or model parameters may be utilized by a user of the workstation 13, and of the application 84 in particular, to evaluate performance, identify actual or potential non-linearities and degradations in the process, identify underperforming processes and abnormal process conditions, and recommend tuning adjustments (e.g., tuning parameters). The virtual controller may be used to test the tuning adjustments to predict process performance and analyze the prediction before implementing the tuning adjustments." Paragraph 0055)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 12 (re-numbered):

Landau does not explicitly describe disturbance as described below. However, Mehta teaches the disturbance as described below.

The cited prior art describes the computer-implemented method of claim 8, wherein the closed loop control includes a feedback loop to an output of a command law to make an estimation of an input disturbance. (**Mehta: "Control routines, and the function blocks that implement such routines, have been configured in accordance with a number of control techniques, including PID control, fuzzy logic control, and model-based techniques such as**

a Smith Predictor or Model Predictive control (MPC). In model-based control techniques, the parameters used in the routines to determine the closed loop control response are based on the dynamic process response to changes in the manipulated or measured disturbances serving as inputs to the process. A representation of this response of the process to changes in process inputs may be characterized as a process model. For instance, a first-order parameterized process model may specify values for the gain, dead time, and time constant of a self-regulating process or values for integrating gain and dead time of an integrating process." Paragraph 0006; Landau: see the y(t) control signal output from the RST controller using an explicit predictor as illustrated in figure 2.1b)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 13 (re-numbered):

Landau does not explicitly describe equipment as described below. However, Mehta teaches the equipment as described below.

The cited prior art describes the computer-implemented method of claim 8, wherein the device or system comprises equipment to be controlled in an industrial process or system. **(Mehta: "The field devices 15-22 may be any types of devices, such as sensors, valves, transmitters, positioners, etc., while the I/O cards 26 and 28 may be any types of I/O devices conforming to any desired communication or controller protocol." Paragraph 0031; "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the**

host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 14 (re-numbered):

Landau does not explicitly describe dead time as described below. However, Mehta teaches the dead time as described below.

The cited prior art describes the computer-implemented method of claim 8, wherein the time delay comprises a dead time. **(Mehta: "The process model identification control logic 110 may then invoke the data collection and model identification. Upon receiving the model identification results, the process model identification control logic 110 may analyze the results for allowable ranges for gain, time constant and dead time if the process is a self-regulating process or for integrating gain and dead time if the process is an integrating process, where the allowable ranges may be user configurable. Thereafter, the tuning 114 of the control function block 58 may be set by the model and tuning control logic 106 based on the identified model and selected tuning rules." Paragraph 0065; Landau: see the predictor as illustrated in figures 2.1a; see also the time delay variations as described in section 6)**

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 15 (re-numbered):

Landau does not explicitly describe a sensor as described below. However, Mehta teaches the sensor as described below.

The cited prior art describes the computer-implemented method of claim 8, further comprising: sensing operating characteristic(s) of the device or system using one or more sensors, the sensed operating characteristic(s) being feedback for closed loop control. **(Mehta: "As illustrated by the exploded block 30 of FIG. 1, the controller 11 may include a number of single-loop control routines, illustrated as routines 32 and 34, and, if desired, may implement one or more advanced control loops, illustrated as control loop 36. Each such loop is typically referred to as a control module. The single-loop control routines 32 and 34 are illustrated as performing single loop control using a single-input/single-output fuzzy logic control block and a single-input/single-output PID control block, respectively, connected to appropriate analog input (AI) and analog output (AO) function blocks, which may be associated with process control devices such as valves, with measurement devices such as temperature and pressure transmitters, or with any other device within the process control system 10. The advanced control loop 36 is illustrated as including an advanced control block 38 having inputs communicatively connected to one or more AI function blocks and outputs communicatively connected to one or more AO function blocks, although the inputs and outputs of the advanced control block 38 may be connected to any other desired function blocks or control elements to receive other types of inputs and to provide other types of control outputs. The advanced control block 38 may be any type of model predictive control (MPC) block, neural network modeling or control block, a multi-variable fuzzy logic control block, a real-time-optimizer block, etc. It will be understood that the function blocks illustrated in FIG. 1, including the advanced control block 38, can**

be executed by the controller 11 or, alternatively, can be located in and executed by any other processing device, such as one of the workstations 13 or even one of the field devices 19-22." Paragraph 0034; "The field devices 15-22 may be any types of devices, such as sensors, valves, transmitters, positioners, etc., while the I/O cards 26 and 28 may be any types of I/O devices conforming to any desired communication or controller protocol." Paragraph 0031; "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 16 (re-numbered):

Landau does not explicitly tuning as described below. However, Mehta teaches the tuning as described below.

The cited prior art describes the computer-implemented method of claim 8, further comprising:

- analyzing operation(s) of the device or system to determine if the closed loop control needs to be tuned; and **(Mehta: "Using the above controller 11 and workstation 13 architecture, the performance of the process may be improved by identifying underperforming loops, tune controllers and/or loops, perform testing, monitor performance. For example, the controllers 11, and specifically the function block**

plug-in 64 and model identification module 72, utilize learning algorithms to automatically identify new models based on process changes (either automatic, manual or induced) that trigger the controller 11 to collected the control data (e.g., process input data) and operating condition (e.g., process output data) and calculate a new model or model parameters. The model or model parameters may be utilized by a user of the workstation 13, and of the application 84 in particular, to evaluate performance, identify actual or potential non-linearities and degradations in the process, identify underperforming processes and abnormal process conditions, and recommend tuning adjustments (e.g., tuning parameters). The virtual controller may be used to test the tuning adjustments to predict process performance and analyze the prediction before implementing the tuning adjustments." Paragraph 0055)

- adaptively tuning the closed loop control using the adjustable single parameter if the closed loop control needs to be tuned. **(Mehta: "As mentioned above, the function block plug-in 64 may perform various functions, including data collection 102, as well as model and tuning control 104 to calculate tuning variables for the control function block 58." Paragraph 0057; "In accordance with some embodiments of the disclosed technique, the function block plug-in 64 performs data collection, as well as model and tuning control to calculate tuning variables for the control function block 58 and process model identification control logic to monitor data and conditions for trigger events indicating a process change sufficient for model identification." Paragraph 0037; "Using the above controller 11 and workstation 13 architecture, the performance of the process may be improved by identifying**

underperforming loops, tune controllers and/or loops, perform testing, monitor performance. For example, the controllers 11, and specifically the function block plug-in 64 and model identification module 72, utilize learning algorithms to automatically identify new models based on process changes (either automatic, manual or induced) that trigger the controller 11 to collected the control data (e.g., process input data) and operating condition (e.g., process output data) and calculate a new model or model parameters. The model or model parameters may be utilized by a user of the workstation 13, and of the application 84 in particular, to evaluate performance, identify actual or potential non-linearities and degradations in the process, identify underperforming processes and abnormal process conditions, and recommend tuning adjustments (e.g., tuning parameters). The virtual controller may be used to test the tuning adjustments to predict process performance and analyze the prediction before implementing the tuning adjustments." Paragraph 0055)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 17 (re-numbered):

Landau does not explicitly describe a computer as described below. However, Mehta teaches the computer as described below.

The cited prior art describes the computer-implemented method of claim 1, wherein the operations of receiving and performing are implemented by a controller for an industrial process or system. **(Mehta: "The controller 11 includes a processor 23 that implements or oversees**

one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claim 18 (re-numbered):

Landau does not explicitly describe a computer or tunable as described below. However, Mehta teaches the computer and tunable as described below.

The cited prior art describes a tangible computer medium storing computer executable code, which when executed by one or more processors, is configured to implement a method of controlling a device or system, the method comprising: **(Landau: "The paper investigates the design of robust digital controllers for systems with time delay. A particular class of digital controllers is considered. namely those for which the closed loop poles include the poles of the system to he controlled." Abstract; "Digital controllers offers a natural and easy solution for controlling systems with time delay without restrictions upon the complexity of the plant dynamics." Section 1; Mehta: "The controller 11 includes a processor 23 that implements or oversees one or more process control routines (stored in a memory 24), which may include control loops, stored therein or otherwise associated therewith and communicates with the devices 15-22, the host computers 13 and the data historian 12 to control a process in any desired manner." Paragraph 0032)**

- receiving a setpoint signal as input; and **(Landau: see the y*(t+d+1) input as illustrated in figure 2.1b)**
- performing closed loop control of the device or system by outputting a control signal according to the setpoint signal and a response model of the device or system using a predictive control algorithm, **(Landau: see the y(t) control signal output from the RST controller using an explicit predictor as illustrated in figure 2.1b)**
    - the predictive control algorithm being configured to implement control according to a polynomial representation for regulation, sensitivity and tracking and **(Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)**
    - further implement non-linearity or time delay compensation using the response model, **(Landau: see the predictor as illustrated in figures 2.1a; see also the time delay variations as described in section 6)**
    - the closed loop control being tunable using an adjustable single parameter for accelerating or decelerating the closed loop control relative to an open loop control scenario. **(Mehta: "In accordance with some embodiments of the disclosed technique, the function block plug-in 64 performs data collection, as well as model and tuning control to calculate tuning variables for the control function block 58 and process model identification control logic to monitor data and conditions for trigger events indicating a process change sufficient for model identification." Paragraph 0037; "Using the above controller 11 and workstation 13 architecture, the performance of the process may be improved by identifying underperforming loops, tune**

controllers and/or loops, perform testing, monitor performance. For example, the controllers 11, and specifically the function block plug-in 64 and model identification module 72, utilize learning algorithms to automatically identify new models based on process changes (either automatic, manual or induced) that trigger the controller 11 to collected the control data (e.g., process input data) and operating condition (e.g., process output data) and calculate a new model or model parameters. The model or model parameters may be utilized by a user of the workstation 13, and of the application 84 in particular, to evaluate performance, identify actual or potential non-linearities and degradations in the process, identify underperforming processes and abnormal process conditions, and recommend tuning adjustments (e.g., tuning parameters). The virtual controller may be used to test the tuning adjustments to predict process performance and analyze the prediction before implementing the tuning adjustments." Paragraph 0055)

Landau and Mehta are combinable for the same rationale as set forth above with respect to claim 1.

Claims 2 and 10 are rejected under 35 U.S.C. 103 as being unpatentable over I. D. Landau, "Robust digital control of systems with time delay (the Smith predictor revisited)," Proceedings of 1994 33rd IEEE Conference on Decision and Control, Lake Buena Vista, FL, USA, 1994, pp. 865-869 vol.1 (Landau) (cited by

Applicant) in view of U.S. Patent Application Publication No. 2009/0112335 (Mehta) and further in view of A. Jacknoon, M. Hassan and S. E. Ferik, "Design of RST controllers based on intelligent optimization algorithms," 2016 Conference of Basic Sciences and Engineering Studies (SGCAC), Khartoum, Sudan, 2016, pp. 177-182 (Jacknoon).

Claim 2:

The cited prior art describes the control system of claim 1,

- wherein the polynomial representation comprises at least transfer functions R, S and T for regulation, sensitivity and tracking, respectively, and **(Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)**

Landau and Mehta do not explicitly describe coefficients as described below. However, Jacknoon teaches the coefficients as described below.

- the coefficients for the transfer functions R, S and T are a function of the adjustable single parameter. **(Jacknoon: see the tuning techniques for the RST structure as illustrated in Tables 1, 2 and as described in section III; "In this paper we present an approach for designing and tuning the RST controller by using intelligence optimization algorithms. Three algorithms namely (BSOA, DE and PSO) have been tested in tuning RST controller that control electrical DC motor that has been used as a benchmark. Simulation results show PSO tuned RST structure to be more**

robust and have high performance." Abstract; Mehta: "In accordance with some embodiments of the disclosed technique, the function block plug-in 64 performs data collection, as well as model and tuning control to calculate tuning variables for the control function block 58 and process model identification control logic to monitor data and conditions for trigger events indicating a process change sufficient for model identification." Paragraph 0037; "Using the above controller 11 and workstation 13 architecture, the performance of the process may be improved by identifying underperforming loops, tune controllers and/or loops, perform testing, monitor performance. For example, the controllers 11, and specifically the function block plug-in 64 and model identification module 72, utilize learning algorithms to automatically identify new models based on process changes (either automatic, manual or induced) that trigger the controller 11 to collected the control data (e.g., process input data) and operating condition (e.g., process output data) and calculate a new model or model parameters. The model or model parameters may be utilized by a user of the workstation 13, and of the application 84 in particular, to evaluate performance, identify actual or potential non-linearities and degradations in the process, identify underperforming processes and abnormal process conditions, and recommend tuning adjustments (e.g., tuning parameters). The virtual controller may be used to test the tuning adjustments to predict process performance and analyze the prediction before implementing the tuning adjustments." Paragraph 0055; Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)

One of ordinary skill in the art would have recognized that applying the known technique of Landau, namely, a digital control system, and the known techniques of Mehta, namely, intelligent control and monitoring in a process control system, with the known techniques of Jacknoon, namely, RST controllers with intelligent optimization algorithms, would have yielded predictable results and resulted in an improved system. Accordingly, applying the teachings of Landau to use RST with a predictor for a digital control system and the teachings of Mehta to use various optimization techniques for a process control system with the teachings of Jacknoon to optimize components of RST controllers would have been recognized by those of ordinary skill in the art as resulting in an improved process control system (i.e., using various tuning and optimization techniques for a RST with a predictor digital control system of Landau based on the teachings of using tuning techniques for a process control system in Mehta and the teachings of using optimization techniques for RST controllers in Jacknoon).

Claim 10 (re-numbered):

The cited prior art describes the computer-implemented method of claim 8,

- wherein the polynomial representation comprises at least transfer functions for R, S and T for regulation, sensitivity and tracking, respectively, **(Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)**

Landau and Mehta do not explicitly describe coefficients as described below. However, Jacknoon teaches the coefficients as described below.

- a coefficient of each of the transfer functions R, S and T being a function of the adjustable single parameter. **(Jacknoon: see the tuning techniques for the RST structure as illustrated in Tables 1, 2 and as described in section III; "In this paper we present an approach for designing and tuning the RST controller by using intelligence optimization algorithms. Three algorithms namely (BSOA, DE and PSO) have been tested in tuning RST controller that control electrical DC motor that has been used as a benchmark. Simulation results show PSO tuned RST structure to be more robust and have high performance." Abstract; Mehta: "In accordance with some embodiments of the disclosed technique, the function block plug-in 64 performs data collection, as well as model and tuning control to calculate tuning variables for the control function block 58 and process model identification control logic to monitor data and conditions for trigger events indicating a process change sufficient for model identification." Paragraph 0037; "Using the above controller 11 and workstation 13 architecture, the performance of the process may be improved by identifying underperforming loops, tune controllers and/or loops, perform testing, monitor performance. For example, the controllers 11, and specifically the function block plug-in 64 and model identification module 72, utilize learning algorithms to automatically identify new models based on process changes (either automatic, manual or induced) that trigger the controller 11 to collected the control data (e.g., process input data) and operating condition (e.g., process output data) and calculate a new model or model parameters. The model or model parameters may be utilized by a user of the workstation 13, and of the application 84 in particular, to evaluate performance, identify actual or potential non-linearities**

and degradations in the process, identify underperforming processes and abnormal process conditions, and recommend tuning adjustments (e.g., tuning parameters). The virtual controller may be used to test the tuning adjustments to predict process performance and analyze the prediction before implementing the tuning adjustments." Paragraph 0055; Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)

Landau, Mehta, and Jacknoon are combinable for the same rationale as set forth above with respect to claim 2.

Claims 3 and 11 are rejected under 35 U.S.C. 103 as being unpatentable over I. D. Landau, "Robust digital control of systems with time delay (the Smith predictor revisited)," Proceedings of 1994 33rd IEEE Conference on Decision and Control, Lake Buena Vista, FL, USA, 1994, pp. 865-869 vol.1 (Landau) (cited by Applicant) in view of U.S. Patent Application Publication No. 2009/0112335 (Mehta) and further in view of U.S. Patent Application Publication No. 2009/0043546 (Srinivasan).

Claim 3:

Landau and Mehta do not explicitly describe linear and non-linear as described below. However, Srinivasan teaches the linear and non-linear as described below.

The cited prior art describes the control system of claim 1,

- wherein the response model includes a linear part and a non-linear part, and **(Srinivasan: "In one exemplary embodiment of the present disclosure, a method of controlling a non-linear process is provided. The method can include providing a non-linear model based at least in part on the non-linear process; providing a plurality of linearized models based at least in part on the non-linear model, where the plurality of linearized models are linearized at different linearization rates; selecting one of the plurality of linearized models based at least in part on a comparison of the plurality of linearized models with a reference model; and controlling the non-linear process by sending one or more control signals from a controller to at least one control device. The one or more control signals can be determined using the selected one of the plurality of linearized models." Paragraph 0010; Landau: see the predictor as illustrated in figures 2.1a; see also the time delay variations as described in section 6)**
- the predictive control algorithm employs the response model to remove non- linearity, thereby compensating for time delay and enabling linear control through an RST control strategy. **(Srinivasan: "In another exemplary embodiment, a control system operably coupled to a processing system is provided. The control system can include one or more control devices operably coupled to a processing system for controlling a process of the processing system; a modeling tool to provide a non-linear model based at least in part on the process and to provide a plurality of linearized models based at least in part on the non-linear model, where the plurality of linearized models are linearized at different linearization rates; and a controller operably**

coupled to the modeling tool. The controller can select one of the plurality of linearized models based on a comparison of the plurality of linearized models with a reference model. The controller can send one or more control signals to at least one of the one or more control devices, and the one or more control signals can be determined using the selected one of the plurality of linearized models." Paragraph 0011; "The present disclosure also contemplates the use of other components, and combinations of components that can receive and/or retrieve operational parameters; retrieve, receive and/or generate non-linear models; retrieve, receive and/or generate linearized models from the non-linear models; and/or implement control with respect to one or more processes or systems." Paragraph 0022; "The modeling tool 75 can have a linearizer 80 to provide or otherwise generate linearized models (e.g., a state space model), which can include piecewise linear models, based upon a generally non-linear model of the non-linear process, at the request or query from the controller 25." Paragraph 0027; Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)

One of ordinary skill in the art would have recognized that applying the known technique of Landau, namely, a digital control system, and the known techniques of Mehta, namely, intelligent control and monitoring in a process control system, with the known techniques of Srinivasan, namely, determining control signals in a process control system, would have yielded predictable results and resulted in an improved system. Accordingly, applying the teachings of Landau to use RST with a predictor for a digital control system and the teachings of Mehta to use various optimization techniques for a process control system with the teachings of Srinivasan to linearize non-linear components of a process control system would have been recognized by

those of ordinary skill in the art as resulting in an improved process control system (i.e., using various tuning and linearization techniques for a RST with a predictor digital control system of Landau based on the teachings of using tuning techniques for a process control system in Mehta and the teachings of using linearization techniques for process controllers in Srinivasan).

Claim 11 (re-numbered):

Landau and Mehta do not explicitly describe linear and non-linear as described below. However, Srinivasan teaches the linear and non-linear as described below.

The cited prior art describes the computer-implemented method of claim 8,

- wherein the response model includes a linear part and a non-linear part, and **(Srinivasan: "In one exemplary embodiment of the present disclosure, a method of controlling a non-linear process is provided. The method can include providing a non-linear model based at least in part on the non-linear process; providing a plurality of linearized models based at least in part on the non-linear model, where the plurality of linearized models are linearized at different linearization rates; selecting one of the plurality of linearized models based at least in part on a comparison of the plurality of linearized models with a reference model; and controlling the non-linear process by sending one or more control signals from a controller to at least one control device. The one or more control signals can be determined using the selected one of the plurality of linearized models." Paragraph 0010; Landau: see the predictor as illustrated in figures 2.1a; see also the time delay variations as described in section 6)**

- the predictive control algorithm employs the response model to remove non-linearity, thereby compensating for time delay and enabling linear control through an RST control strategy. (**Srinivasan: "In another exemplary embodiment, a control system operably coupled to a processing system is provided. The control system can include one or more control devices operably coupled to a processing system for controlling a process of the processing system; a modeling tool to provide a non-linear model based at least in part on the process and to provide a plurality of linearized models based at least in part on the non-linear model, where the plurality of linearized models are linearized at different linearization rates; and a controller operably coupled to the modeling tool. The controller can select one of the plurality of linearized models based on a comparison of the plurality of linearized models with a reference model. The controller can send one or more control signals to at least one of the one or more control devices, and the one or more control signals can be determined using the selected one of the plurality of linearized models." Paragraph 0011; "The present disclosure also contemplates the use of other components, and combinations of components that can receive and/or retrieve operational parameters; retrieve, receive and/or generate non-linear models; retrieve, receive and/or generate linearized models from the non-linear models; and/or implement control with respect to one or more processes or systems." Paragraph 0022; "The modeling tool 75 can have a linearizer 80 to provide or otherwise generate linearized models (e.g., a state space model), which can include piecewise linear models, based upon a generally non-linear model of the non-linear process, at the request or query

from the controller 25." Paragraph 0027; Landau: see the T, S, and R as illustrated in figure 2.1a; see also the T, R, and S polynomials as illustrated in sections 2, 4)

Landau, Mehta, and Srinivasan are combinable for the same rationale as set forth above with respect to claim 3.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

- I.D. Landau, The R-S-T digital controller design and applications, Control Engineering Practice, Volume 6, Issue 2, 1998, Pages 155-165 describes a RST digital controller.
- O'Dwyer, Aidan : A reference guide to Smith predictor based methods for the compensation of dead-time processes. Proceedings of the IEE Irish Signals and Systems Conference, pp. 231-238, Dublin City University, September, 2005. doi:10.21427/6sck-s48 describes using a Smith predictor for compensation of dead-time processes.
- Ioan D. Landau and Gianluca Zito, DIGITAL CONTROL SYSTEMS Identification, design and implementation, GIPSA – LAB, 2020 describes various digital controller mechanisms including RST controllers.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CHRISTOPHER E EVERETT whose telephone number is

(571)272-2851. The examiner can normally be reached Monday-Friday 8:00 am to 5:00 pm (Eastern).

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kenneth Lo can be reached on 571-272-9774. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Christopher E. Everett/
Primary Examiner, Art Unit 2116